STRICT TRANSFORMS PRESERVE UNIVERSALLY EXACT SEQUENCES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The height of this card is H2 rather than H3 because the geometric input is reduced to a single algebra lemma about a-power torsion, which the author proves in full.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Reduction of the strict transform to the affine case. Lemma lemma-strict-transform-affine, used at proof lines 9052-9055. Allows pushing the sheaves to S and working with modules.
- Flat base change of a blowup reduces to an effective Cartier divisor. used at proof lines 9055-9058. Turns the blowup into the localisation A -> A[I/a].
- Universal injectivity of M_1 -> M_2. used at proof lines 9066-9069. Is the hypothesis that closes the exactness in the middle.

Source reference. Original definition 080D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10178–10178. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 080G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10181–10181. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *In the situation of Definition 080D. Suppose that*

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$$

is an exact sequence of quasi-coherent sheaves on X which remains exact after any base change $T \to S$. Then the strict transforms of $\mathcal{F}_i'$ relative to any blowup $S' \to S$ form a short exact sequence $0 \to \mathcal{F}_1' \to \mathcal{F}_2' \to \mathcal{F}_3' \to 0$ too.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We may localize on S and X and assume both are affine. Then we may push $\mathcal{F}_i$ to S, see Lemma 080G. We may assume that our blowup is the morphism $1 : S \to S$ associated to an effective Cartier divisor $D \subset S$. Then the translation into algebra is the following: Suppose that A is a ring and $0 \to M_1 \to M_2 \to M_3 \to 0$ is a universally exact sequence of A-modules. Let $a \in A$. Then the sequence

$$0 \to M_1/a\text{-power torsion} \to M_2/a\text{-power torsion} \to M_3/a\text{-power torsion} \to 0$$

is exact too. Namely, surjectivity of the last map and injectivity of the first map are immediate. The problem is exactness in the middle. Suppose that $x \in M_2$ maps to zero in M_3/a-power torsion. Then $y = a^n x \in M_1$ for some n. Then y maps to zero in $M_2/a^n M_2$. Since $M_1 \to M_2$ is universally injective we see that y maps to zero in $M_1/a^n M_1$. Thus $y = a^n z$ for some $z \in M_1$. Thus $a^n(x - y) = 0$. Hence y maps to the class of x in M_2/a-power torsion as desired. $\square$